



МИНОБРНАУКИ РОССИИ

**федеральное государственное автономное образовательное учреждение
высшего образования**

«Московский государственный технологический университет

«СТАНКИН»

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Кафедра компьютерных систем управления**

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обеспечения»**

Отчет

**по лабораторной работе №1
«Работа с GitHub»**

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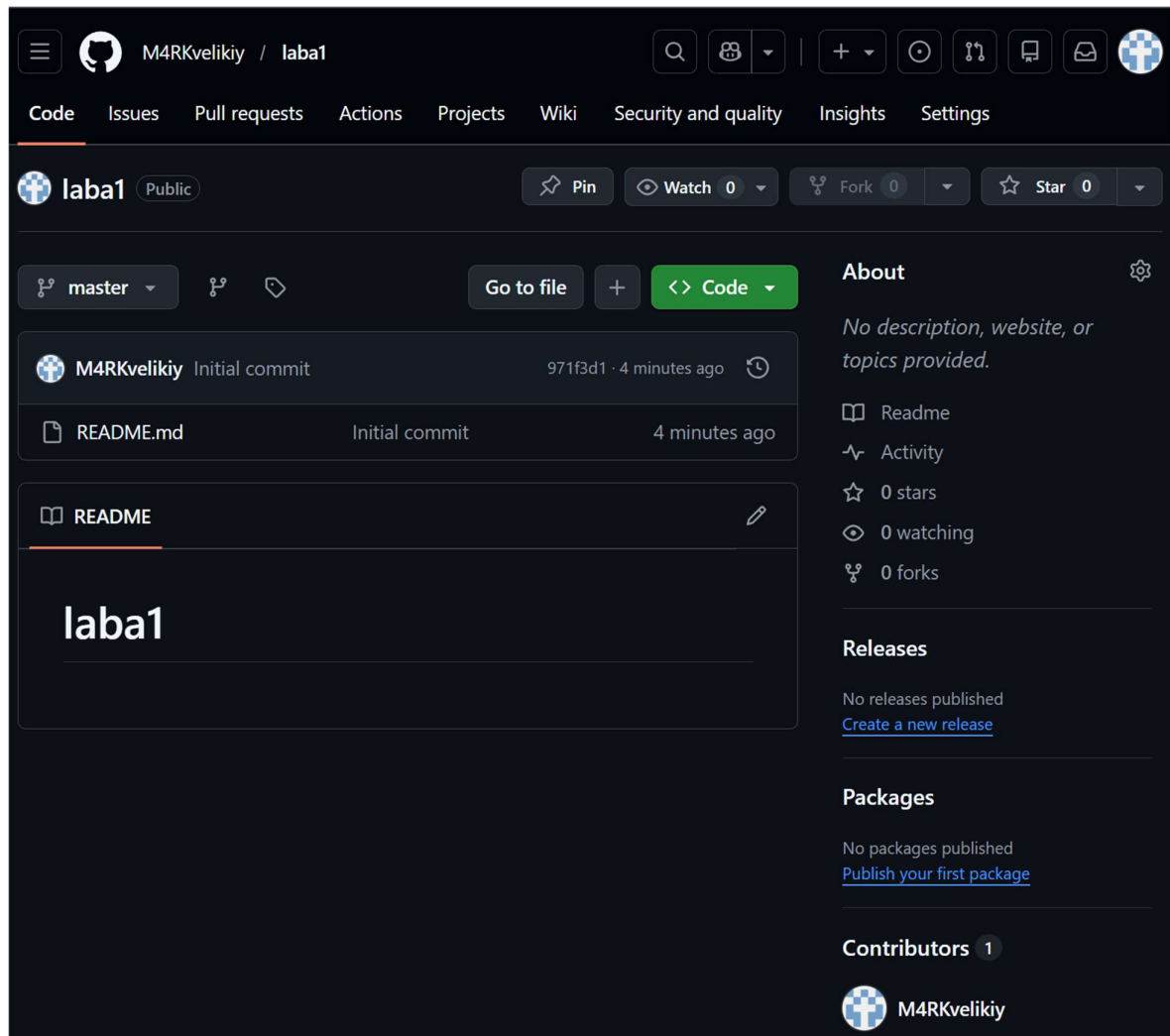
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Цель работы: освоить контроль версий: загрузка и скачивание файлов в репозиторий, управление ветками, разрешение конфликтов и взаимодействие со сторонними репозиториями.

Гит: <https://github.com/M4RKvelikiy/lab1>

Создали аккаунт, поменяли main на master, создали репозиторий



Работа с консолью

Создали папку для нашего локального репозитория, далее прописали свои данные от гита, проинициализировали его в папке:

```
C:\lab1>git config --global user.name "M4RKvelikiy"

C:\lab1>git config --global user.email mark.shedko@gmail.com

C:\lab1>git init
Initialized empty Git repository in C:/lab1/.git/

C:\lab1>
```

В папке создали txt файл с фамилией, посмотрели статус:

```
C:\lab1>git status
On branch master

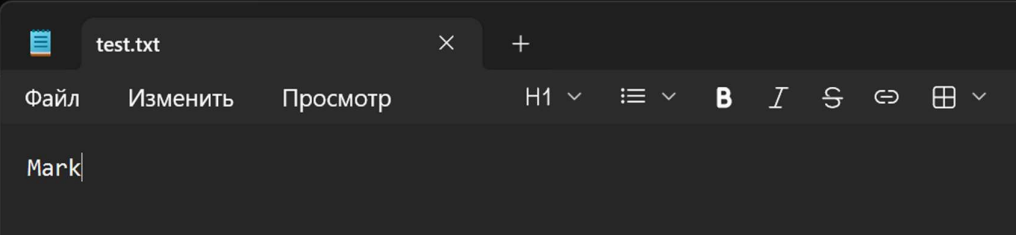
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\lab1>git add .

C:\lab1>git commit -m "first commit"
[master (root-commit) 89ddcd8] first commit
 1 file changed, 1 insertion(+)
 create mode 100644 test.txt
```

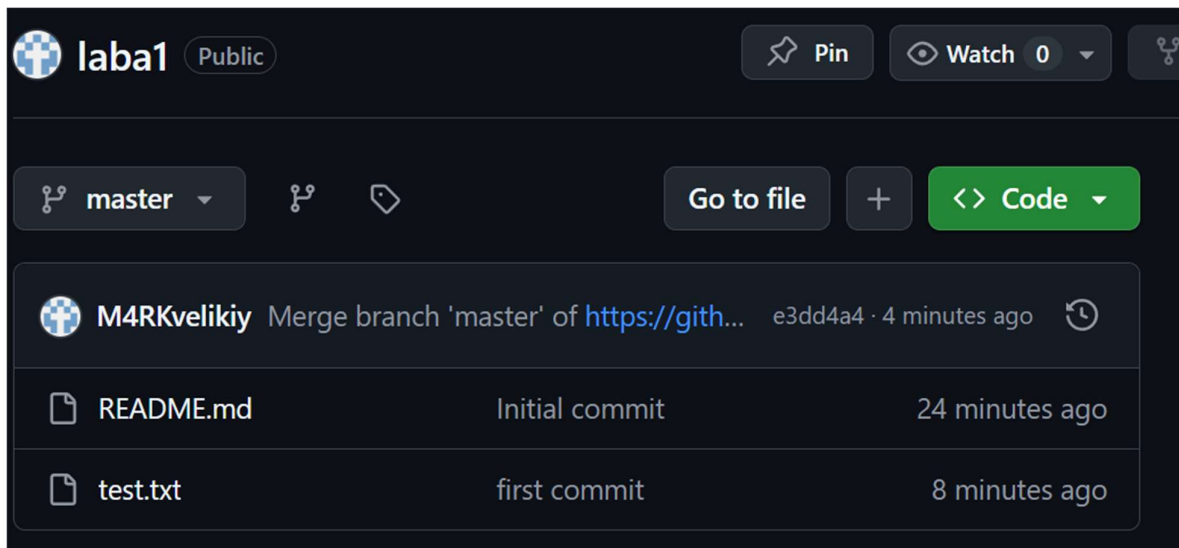


Пулл с сервера

```
C:\lab1>git pull origin master --allow-unrelated-histories
From https://github.com/M4RKvelikiy/lab1
 * branch                master      -> FETCH_HEAD
Merge made by the 'ort' strategy.
 README.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
```

Пуш на сервер

```
C:\lab1>git push origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 524 bytes | 524.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/M4RKvelikiy/lab1
 971f3d1..e3dd4a4  master -> master
```



Лог

```
C:\lab1>git log
commit e3dd4a46fa38778f5fdb722d3a5c8467e8643cba (HEAD -> master, origin/master, origin/HEAD)
Merge: cc1c934 971f3d1
Author: M4RKvelikiy <mark.shedko@gmail.com>
Date: Sun May 17 19:54:04 2026 +0300

    Merge branch 'master' of https://github.com/M4RKvelikiy/lab1

commit cc1c93440357b1afdaa094c8f3b92f6b7fb2a6c7
Author: M4RKvelikiy <mark.shedko@gmail.com>
Date: Sun May 17 19:50:14 2026 +0300

    first commit

commit 971f3d1d5c3bdf84d42ef323a65a2b625c20c518
Author: M4RKvelikiy <mark.shedko@gmail.com>
Date: Sun May 17 19:33:57 2026 +0300
```

Новая ветка и пулл в неё с новым файлом:

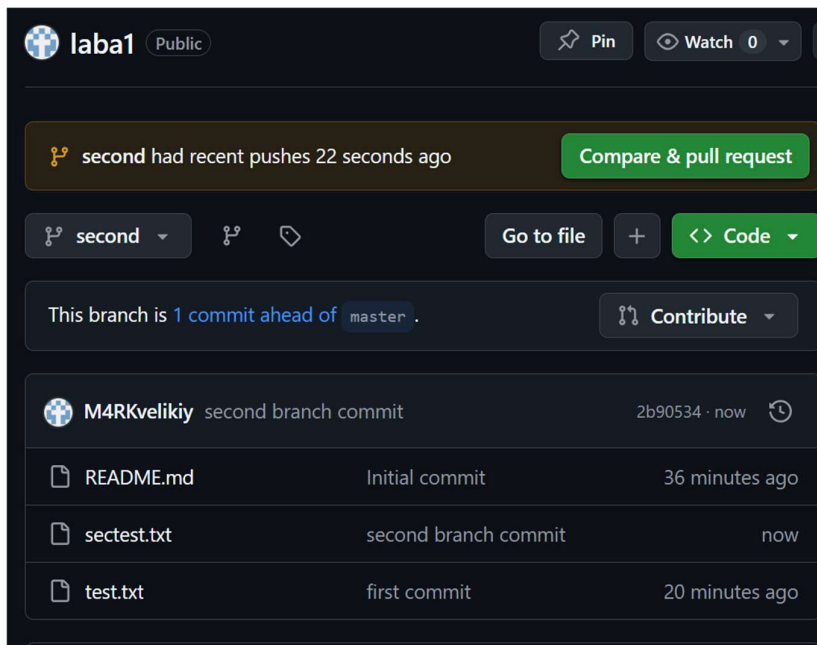
```
C:\lab1>git branch second

C:\lab1>git checkout second
Switched to branch 'second'

C:\lab1>git add .
```

```
C:\lab1>git commit -m "second branch commit"
[second 2b90534] second branch commit
1 file changed, 1 insertion(+)
create mode 100644 sectest.txt

C:\lab1>git push origin second
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 322 bytes | 322.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/M4RKvelikiy/lab1
e3dd4a4..2b90534 second -> second
```



Слияние веток

```
C:\lab1>git checkout master
Switched to branch 'master'

C:\lab1>git merge second
Updating e3dd4a4..3fbfb54
Fast-forward
 sectest.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 sectest.txt

C:\lab1>git branch -d second
Deleted branch second (was 3fbfb54).
```

Ветка newdev

```
C:\lab1>git branch newdev

C:\lab1>git checkout newdev
Switched to branch 'newdev'

C:\lab1>git add .
```

```
C:\lab1>git commit -m "newdev commit"
[newdev 9bc7c01] newdev commit
2 files changed, 1 insertion(+), 2 deletions(-)
delete mode 100644 newdev.txt
```

test.txt × sectest.txt

Файл Изменить Просмотр H1 ▾ ≡ ▾

Mark Antonovich

Переключились на master:

```
C:\lab1>git checkout master
Switched to branch 'master'
```

test.txt × sectes

Файл Изменить Просмотр

Mark

```
C:\lab1>git merge newdev
Updating 3fbfb54..9bc7c01
Fast-forward
 test.txt | 2 + -
1 file changed, 1 insertion(+), 1 deletion(-)
```

test.txt × sectest.txt

Файл Изменить Просмотр H1 ▾ ≡ ▾

Mark Antonovich

Создали конфликт файлов

```
C:\lab1>git checkout newdev
Switched to branch 'newdev'

C:\lab1>git add .

C:\lab1>git commit -m "err commit"
[newdev 1a20677] err commit
1 file changed, 1 insertion(+), 1 deletion(-)

C:\lab1>git checkout master
Switched to branch 'master'

C:\lab1>git add .

C:\lab1>git commit -m "err commit"
[master 4e97c95] err commit
1 file changed, 1 insertion(+), 1 deletion(-)

C:\lab1>git merge newdev
Auto-merging test.txt
CONFLICT (content): Merge conflict in test.txt
Automatic merge failed; fix conflicts and then commit the result.
```

```
C:\lab1>git diff
diff --cc test.txt
index 6635b0e,37f6bc5..0000000
--- a/test.txt
+++ b/test.txt
@@@ -1,1 -1,1 +1,5 @@@
- Mark Antonovichhhh
- Mark Antonov
++<<<<<< HEAD
++Mark Antonovichhhh
++=====
++Mark Antonov
++>>>>>> newdev
```

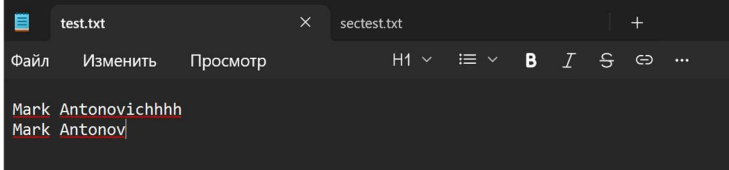
Пофиксили и отправили на сервер, newdev удалили

```
C:\lab1>git add .

C:\lab1>git commit -m "err fix commit"
[master a41afa6] err fix commit

C:\lab1>git push origin master
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 16 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (14/14), 1.21 KiB | 309.00 KiB/s, done.
Total 14 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 1 local object.
To https://github.com/M4RKvelikiy/lab1
e3dd4a4..a41afa6 master -> master

C:\lab1>git branch -d newdev
Deleted branch newdev (was 1a20677).
```



Удаляем на сервере(ошибка, потому что newdev был локально, на сервер не пушили)

```
C:\lab1>git push origin --delete newdev
error: unable to delete 'newdev': remote ref does not exist
error: failed to push some refs to 'https://github.com/M4RKvelikiy/lab1'

C:\lab1>git push origin --delete second
To https://github.com/M4RKvelikiy/lab1
- [deleted]          second
```

Возврат в предыдущее состояние

```
C:\lab1>git log
commit a41afa63d4c5f66ac471981bd698051510414028 (HEAD -> master, origin/master, origin/HEAD)
Merge: 4e97c95 1a20677
Author: M4RKvelikiy <mark.shedko@gmail.com>
Date: Sun May 17 20:25:22 2026 +0300

    err fix commit

commit 4e97c95871f6441bc82fe9015a7d94e74c9f437a
Author: M4RKvelikiy <mark.shedko@gmail.com>
Date: Sun May 17 20:23:26 2026 +0300

    err commit
```

```
C:\lab1>git checkout 4e97c9587
Note: switching to '4e97c9587'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

    git switch -c <new-branch-name>

Or undo this operation with:

    git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 4e97c95 err commit

C:\lab1>git branch
* (HEAD detached at 4e97c95)
  master
```

Исправление коммита:

```
C:\lab1>git branch
* master

C:\lab1>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   sectest.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\lab1>git add .

C:\lab1>git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   sectest.txt

C:\lab1>git reset HEAD
Unstaged changes after reset:
M       sectest.txt

C:\lab1>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   sectest.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

```
C:\lab1>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   sectest.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\lab1>git add .

C:\lab1>git commit -m "reset commit"
[master 6bbdd82] reset commit
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\lab1>git reset HEAD

C:\lab1>git status
On branch master
nothing to commit, working tree clean

C:\lab1>
```

Изменение комментария git commit –amend

```
reset commit
```

```
# Please enter the commit message for your changes. Lines starting  
# with '#' will be ignored, and an empty message aborts the commit.
```

```
#
```

```
# Date:      Sun May 17 20:34:16 2026 +0300
```

```
#
```

```
# On branch master
```

```
# Changes to be committed:
```

```
#       modified:   sectest.txt
```

```
#
```

```
~
```

```
~
```

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```

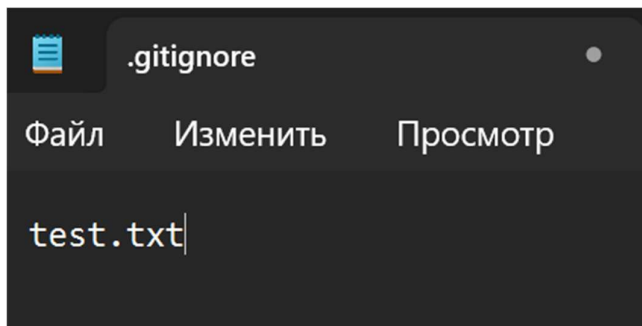
```
~
```

```
~
```

```
~
```

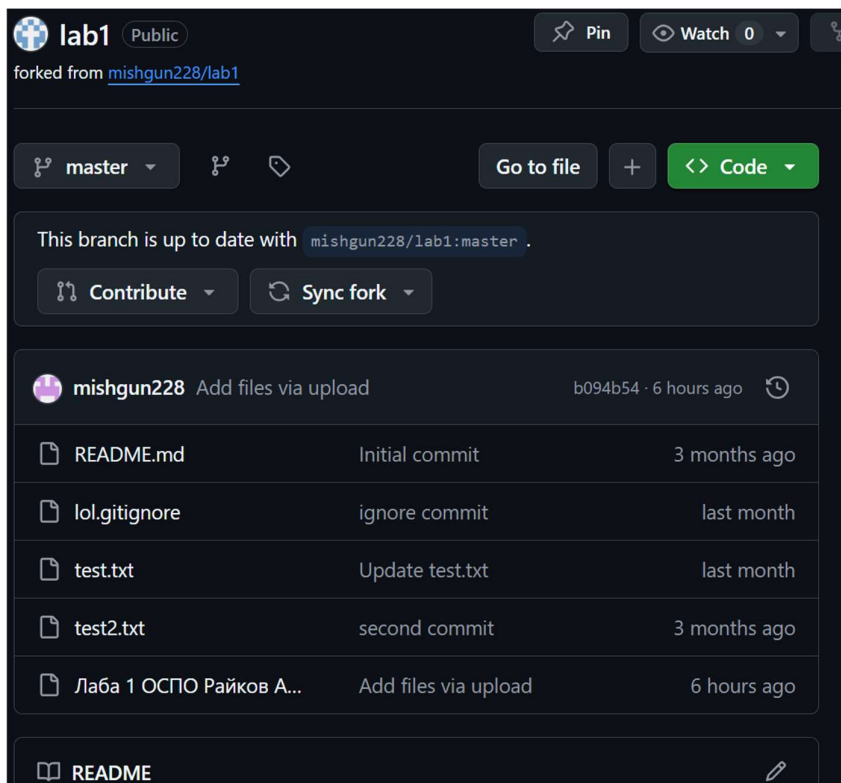
```
.git/COMMIT_EDITMSG [unix] (20:35 17/05/2026)
```

```
1,1 AL
```

Совместная работа

Делаем форк репозитория товарища:



Клонируем:

```
C:\lab1>git clone https://github.com/mishgun228/lab1
Cloning into 'lab1'...
remote: Enumerating objects: 61, done.
remote: Counting objects: 100% (61/61), done.
remote: Compressing objects: 100% (42/42), done.
remote: Total 61 (delta 9), reused 48 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (61/61), 2.39 MiB | 954.00 KiB/s, done.
Resolving deltas: 100% (9/9), done.
```

Просим дать доступ к репозиторию и начинаем работу:

Пуш изменённого файла:

```
C:\lab1\lab1>git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test2.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\lab1\lab1>git add .

C:\lab1\lab1>git commit -m "shedko commit"
[master 7816ab3] shedko commit
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\lab1\lab1>git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 276 bytes | 276.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mishgun228/lab1
   b094b54..7816ab3  master -> master
```

lab1 / test2.txt 

 M4RKvelikiy shedko commit

Code

Blame

1 lines (1 loc) · 15 Bytes

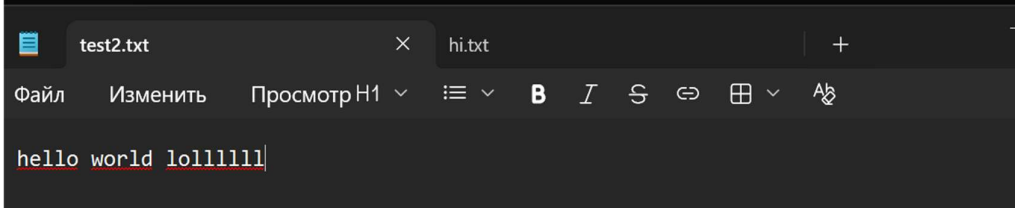


1

hello world lol

Этот же файл поменяли, запулим на локальный сервер

```
C:\lab1\lab1>git pull origin master
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 921 bytes | 28.00 KiB/s, done.
From https://github.com/mishgun228/lab1
 * branch                master      -> FETCH_HEAD
   c447dd9..5112efc master      -> origin/master
Updating c447dd9..5112efc
Fast-forward
 test2.txt | 2 +
 1 file changed, 1 insertion(+), 1 deletion(-)
```



Пуш нового файла:

```
C:\lab1\lab1>git status
On branch master
Your branch is up to date with 'origin/master'.

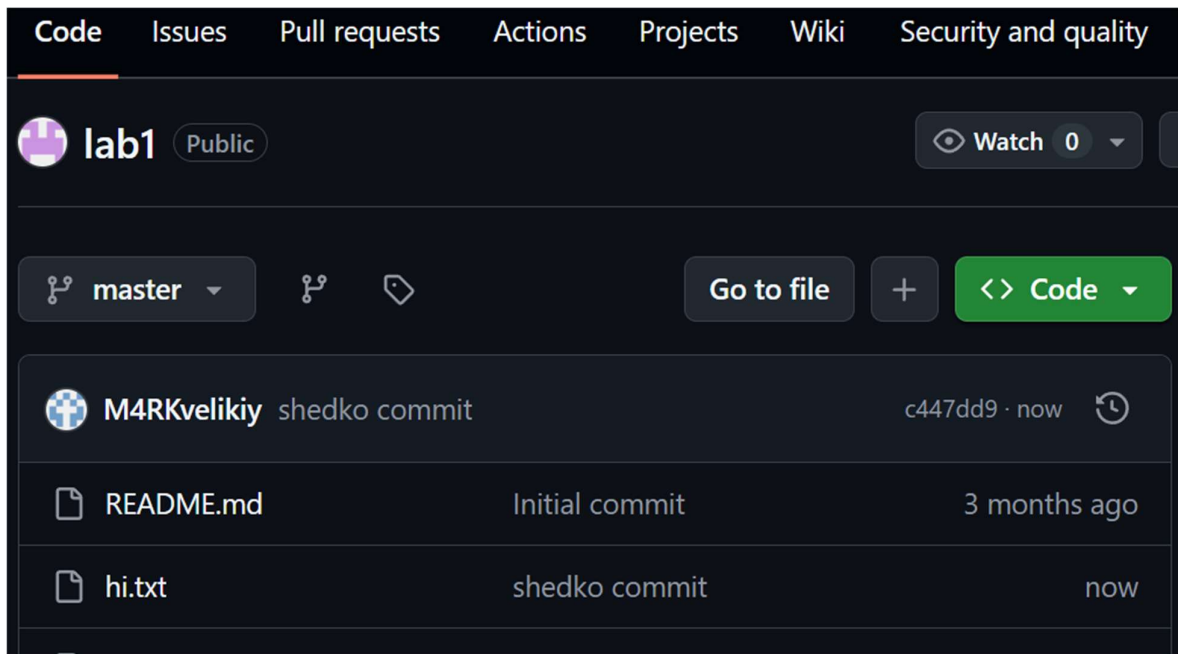
Untracked files:
  (use "git add <file>..." to include in what will be committed)
      hi.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\lab1\lab1>git add .

C:\lab1\lab1>git commit -m "shedko commit"
[master c447dd9] shedko commit
 1 file changed, 1 insertion(+)
 create mode 100644 hi.txt

C:\lab1\lab1>git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 271 bytes | 271.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mishgun228/lab1
 7816ab3..c447dd9 master -> master
```

Получили ошибку при редактировании одного и того же файла:

```
C:\lab1\lab1>git push origin master
To https://github.com/mishgun228/lab1
 ! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'https://github.com/mishgun228/lab1'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

C:\lab1\lab1>git pull origin master
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 917 bytes | 53.00 KiB/s, done.
From https://github.com/mishgun228/lab1
 * branch                master      -> FETCH_HEAD
  5112efc..03193f5  master      -> origin/master
Auto-merging test2.txt
CONFLICT (content): Merge conflict in test2.txt
Automatic merge failed; fix conflicts and then commit the result.

C:\lab1\lab1>git diff
diff --cc test2.txt
index 2300cfb,132f8a4..0000000
--- a/test2.txt
+++ b/test2.txt
@@@ -1,1 -1,1 +1,5 @@@
++<<<<<< HEAD
+hello world lolll
++=====
+ hello world lol
++>>>>>> 03193f5b724ff59c76663f3e07de395dd8663bde
```


Вывод:

В процессе лабораторной работы были получены практические навыки использования Git и GitHub: инициализация репозитория, фиксация версий (коммиты), взаимодействие с удалённым хранилищем, работа с ветками, слияние и разрешение конфликтов, откат к предыдущим состояниям, применение файла .gitignore, а также базовые приёмы совместной разработки.

Список литературы

1. Git - Documentation : официальный сайт. – URL: <https://git-scm.com/doc> (дата обращения: 04.05.2026). – Текст : электронный.
2. GitHub Docs : официальный сайт. – URL: <https://docs.github.com> (дата обращения: 04.05.2026). – Текст : электронный.
3. Git - Downloads : официальный сайт. – URL: <https://git-scm.com/downloads> (дата обращения: 04.05.2026). – Текст : электронный.
4. Learn Git Branching : интерактивный обучающий ресурс. – URL: https://learngitbranching.js.org/?locale=ru_RU (дата обращения: 04.05.2026). – Изображение. Устная речь : электронные.
5. Чакон, С. Pro Git / С. Чакон, Б. Штрауб. – 2-е изд. – URL: <https://git-scm.com/book/ru/v2> (дата обращения: 04.05.2026). – Текст : электронный.